

# SOWGOTO RAHA SUNNY

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## Academic Profile

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M.S. graduate in Cybersecurity with a GPA of 3.72/4.00 and research experience in machine and deep learning, explainable and trustworthy AI, anomaly detection, network security, and privacy-preserving federated learning. Research interests include AI-driven cybersecurity, secure and reliable intelligent systems, large language models (LLMs), and distributed AI. Seeking Ph.D. opportunities to develop robust, interpretable, and privacy-aware learning methods for secure networked systems, edge intelligence, and autonomous systems.

## Education

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| <b>Master of Science in Cybersecurity</b><br><i>University of North Texas, Denton, TX</i> | 2026<br>GPA: 3.72/4.00 |
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| <b>Master of Science in Information Technology</b><br><i>Jahangirnagar University, Dhaka, Bangladesh</i> | 2019<br>GPA: 3.01/4.00 |
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| <b>Bachelor of Science in Information Technology</b><br><i>Jahangirnagar University, Dhaka, Bangladesh</i> | 2017<br>GPA: 3.45/4.00 |
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## Research Interests

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Trustworthy and secure artificial intelligence, machine learning and deep learning, explainable AI, anomaly detection, privacy-preserving and distributed learning, reliable edge intelligence, autonomous multi-agent systems, and the security and resilience of intelligent networked systems.

## Research Experience

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| <b>Explainable Deep Learning for Network Intrusion Detection</b><br><i>Independent Research</i> | 2025 – Present<br>Texas, USA |
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- Developed NML-XAI combining Transformer-based feature processing, CNN-BiLSTM-attention-autoencoder models, and ensemble stacking for explainable network intrusion detection.
- Applied SMOTE, SHAP, and LIME to UNSW-NB15 and achieved 93.13% accuracy, 93.45% precision, and 93.19% F1-score.
- Investigated interpretable machine-learning techniques for identifying anomalous network behavior and improving confidence in automated cybersecurity decisions.

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| <b>Hybrid KAN and Explainable AI for DDoS Detection</b><br><i>Collaborative Research Project</i> | 2025 – 2026<br>Texas, USA |
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- Designed a hybrid Kolmogorov–Arnold Network and explainable AI pipeline for multi-class DDoS detection in software-defined networks.
- Evaluated predictive performance and interpretable feature behavior to improve reliable security decisions in networked systems.
- Contributed to experimental design, model evaluation, technical analysis, manuscript preparation, and research documentation.

## Publications and Manuscripts

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1. **S. R. Sunny**, Z. A. Mamun, “HKAN-XAI: A Hybrid Kolmogorov–Arnold Network with Explainable AI for Multi-Class DDoS Attack Detection in SDN,” accepted for oral presentation at the *2nd IEEE International Conference on Cybersecurity and AI-Based Systems (Cyber-AI 2026)*, Bucharest, Romania, 2026.
2. **S. R. Sunny**, S. S. Adhimulam, and P. R. Ovi, “The Future of Database Resilience: A Systematic Literature Review on Mitigating AI-Driven Ransomware in Transactional Systems,” manuscript under review.
3. M. Islam, T. Mohiuddin, M. A. Rahman, Md. Sharfruddin, E. Lokesh, and **S. R. Sunny**, “A Hybrid Deep Learning Framework for Privacy-Preserving Sepsis Prediction in Distributed ICU Environments Using Federated Learning Simulation,” *Discover Social Science and Health*, accepted for publication, 2026.

## Teaching Experience

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| <b>Graduate Teaching Assistant – Applied Machine Learning</b><br><i>University of North Texas</i> | Jan 2025 – May 2025<br>Denton, TX |
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- Supported graduate-level instruction in applied machine learning, programming, and computational problem solving.
- Provided office-hour and laboratory support for Python, software configuration, debugging, code review, and technical assignments.
- Guided students on regression, clustering, decision trees, random forests, SVMs, neural networks, and model evaluation.
- Created technical documentation and explained solutions to students with varied programming and technical backgrounds.

## Professional Engineering Experience

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### Senior System Engineer

*Dotlines Bangladesh, Dhaka, Bangladesh*

Jul 2022 – Jul 2024

- Engineered and administered Linux and Windows production environments supporting servers, databases, APIs, web services, telecommunications, and transactional systems.
- Maintained 99.9% service availability across platforms processing more than 50,000 daily transactions through monitoring, troubleshooting, automation, and incident response.
- Developed Python, Bash, SQL, PL/SQL, and PowerShell automation for diagnostics, job monitoring, data analysis, and service-health validation.
- Reduced mean time to resolution by 30% across 35+ monthly incidents using Linux log analysis, database investigation, API validation, and structured root-cause analysis.
- Collaborated with development, DevOps, cybersecurity, database, and infrastructure teams on CI/CD deployments, rollback validation, testing, and production troubleshooting.
- Performed vulnerability assessment, security remediation, access reviews, backup verification, and system-hardening activities.

### System Engineer

*Dotlines Bangladesh, Dhaka, Bangladesh*

May 2019 – Jun 2022

- Supported Linux/Windows servers, databases, enterprise applications, networking, payment gateways, VAS, bulk SMS/OTP, and telecommunications services.
- Resolved 60+ weekly incidents involving application errors, SQL/database issues, Linux services, network connectivity, user access, file transfers, and UAT validation while meeting SLA targets.
- Performed server maintenance, patch validation, backup verification, access reviews, security remediation, and service-health monitoring.
- Created SOPs, troubleshooting guides, runbooks, RCA records, and technical documentation to support engineering and operations teams.

## Selected Research and Technical Projects

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**Ransomware Attack Simulation and Mitigation** — Developed a Python-based monitoring and mitigation environment using Watchdog, psutil, and SQLite logging to study ransomware behavior and defensive controls.

**AI-Enhanced Zero Trust Cybersecurity** — Designed a cloud-oriented Zero Trust architecture incorporating continuous authentication, anomaly detection, least-privilege access, and automated policy enforcement.

**Web Vulnerability Scanner Analysis** — Evaluated OpenVAS, Nikto, and Wapiti against OWASP-relevant targets, comparing vulnerability-detection capability and false-positive behavior.

## Technical Skills

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**Programming & Scripting:** Python, Java, SQL, PL/SQL, Bash/Shell, PowerShell, C/C++

**Machine Learning:** Transformers, CNN, RNN, BiLSTM, autoencoders, ensemble learning, SVM, SMOTE, model evaluation  
**Explainable / Trustworthy AI:** SHAP, LIME, explainability, anomaly detection, uncertainty-aware and reliability-focused analysis

**Systems:** Linux (Ubuntu, RHEL), UNIX, Windows Server, Windows 10/11, macOS

**Cloud & DevOps:** AWS, Microsoft Azure, Docker, Kubernetes, VMware, Hyper-V, Git/GitHub, Jenkins, CI/CD, Terraform, Ansible

**Databases:** MySQL, MariaDB, SQL Server, Oracle, PostgreSQL, ETL pipelines, batch processing

**Networking & Security:** TCP/IP, DNS, DHCP, VPN, Wireshark, vulnerability assessment, network security, access control, incident response

**Teaching:** Python instruction, laboratory support, code review, debugging, student mentoring, technical documentation, instructional technology

## Certification

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**AWS Certified Solutions Architect – Associate | Credential Link**

June 2026